Practice Quiz 08A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 08

Well-definedness on a quotient; onto maps and inverses.

Write $[x]_5$ for the congruence class of the integer x modulo 5, and write $\mathbf{Z}_5$ for the set of these classes. Thus $[x]_5 = [y]_5$ if and only if $x - y = 5k$ for some integer k.

1. Consider the proposed rule

$$f : \mathbf{Z}_5 \longrightarrow \mathbf{Z}_5, \qquad f([x]_5) = [x+3]_5.$$

(a) Prove that f is well defined.

Solution Let $x, y \in \mathbf{Z}$ and suppose $[x]_5 = [y]_5$. Choose an integer k such that $x - y = 5k$. Then

$$(x+3) - (y+3) = x - y = 5k,$$

so $[x+3]_5 = [y+3]_5$. Thus the output does not depend on the representative, and f is well defined. ∎

(b) Prove that f is onto and determine its inverse.

Solution Let $[y]_5 \in \mathbf{Z}_5$ be arbitrary. The element $[y-3]_5$ is a preimage, because $f([y-3]_5) = [(y-3)+3]_5 = [y]_5$. Hence f is onto.

The inverse is $g([y]_5) = [y-3]_5$. This rule is well defined: if $y - z = 5k$, then $(y-3) - (z-3) = 5k$. Moreover,

$$f(g([y]_5)) = [y]_5, \qquad g(f([x]_5)) = [x]_5.$$

Thus both composites are identity functions, so $g = f^{-1}$. ∎

Solutions revised using GPT-6 Astra in the Codex harness.